IAT New Test 23

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

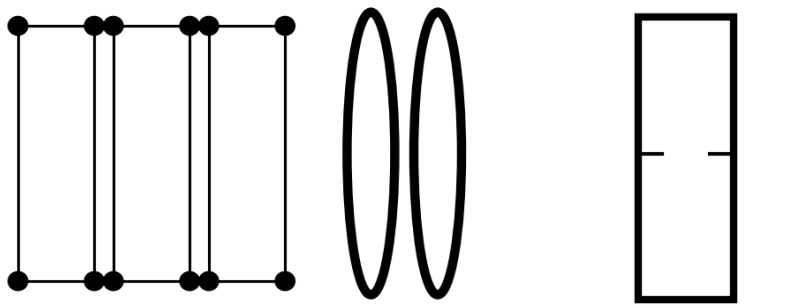
Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A classification key lists P with jointed appendages and an exoskeleton, Q with a mantle and muscular foot, R with an adult water-vascular system and pentaradial body plan, and S with proboscis, collar and trunk. Which ordered assignment fits?
A) Annelida, Mollusca, Cnidaria, Chordata
B) Arthropoda, Annelida, Echinodermata, Chordata
C) Arthropoda, Mollusca, Cnidaria, Hemichordata
D) Arthropoda, Mollusca, Echinodermata, Hemichordata

Q2 The tissue diagram labels P as living elongated cells with uneven corner thickening, Q as dead heavily lignified fibres with narrow lumens, and R as a perforated conducting tube in a vascular bundle. A defect prevents normal lignification but leaves the corner-thickening process intact. Which prediction best distinguishes the tissue effects?

Three tissue structures: schematic



P: living; corner thickening Q: dead; lignified R: perforated tube

Vascular bundle

A) All three must become identical parenchyma immediately
B) P can retain collenchymatous support, while Q support and R's resistance to collapse are compromised
C) P loses all walls, while Q and R are unchanged because they are dead
D) R becomes phloem solely because its walls contain less lignin

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Q3 Four meiotic events are recorded: homologous chromosomes pair, nonsister chromatids exchange corresponding segments, homologues separate while sister centromeres remain joined, and sister chromatids separate. Which ordering and interpretation are correct?
A) Synapsis precedes crossing over; homologues separate in division I and sisters in division II
B) Crossing over precedes synapsis; sister chromatids separate in division I
C) Homologues and sister chromatids always separate together in the first division
D) All four events are ordinary metaphase events of mitosis

Q4 In a C4 leaf, a four-carbon acid normally moves from mesophyll to bundle-sheath cells, releases CO2 there, and leaves a three-carbon compound that returns for acceptor regeneration. A selective block prevents this return while initially leaving decarboxylation and RuBisCO intact. Which early-to-later effect is most consistent?
A) Calvin-cycle reactions must all relocate into mesophyll cells
B) The initial four-carbon acid directly becomes RuBisCO and replaces its gene
C) Some existing acid can still release CO2, but sustained PEP regeneration and the concentrating shuttle decline
D) The block proves that C4 plants never use ATP for acceptor regeneration

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Q5 In a limited endocrine model, parathyroid output is lost while thyroid thyroxine production and its pituitary feedback remain normal. Blood calcium falls and neuromuscular excitability rises. Which interpretation follows without treating all thyroid-region hormones as one signal?
A) Normal thyroxine proves that parathyroid hormone is also normal
B) Calcitonin deficiency is the stated cause, and PTH would lower calcium further
C) TSH directly supplies all missing calcium to the circulation
D) Loss of calcium-raising PTH fits the model; thyroid calcitonin has an opposing calcium-related action distinct from thyroxine

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Q6 Three assisted-reproduction plans are considered. P has a four-cell laboratory embryo but both oviducts are blocked and the uterus can support transfer. Q requires a donor ovum and has a patent oviduct; fertilisation is intended to occur inside her body. R requires a sperm to be injected directly into an oocyte. Which complete plan fits the stated stages and sites?
A) P culture beyond eight blastomeres then intrauterine transfer; Q GIFT; R ICSI
B) P transfer the four-cell embryo into a blocked tube; Q GIFT; R ICSI
C) P culture then intrauterine transfer; Q transfer an already fertilised embryo as the unfertilised donor ovum; R ICSI
D) P culture then intrauterine transfer; Q GIFT; R insemination into the uterus instead of the oocyte

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Q7 In a supplied cell-free translation experiment, a tRNA with anticodon 3-prime UUU 5-prime is artificially charged with valine instead of its usual lysine. Editing is bypassed, and the ribosome accepts it at mRNA codon 5-prime AAA 3-prime. What does this event demonstrate?
A) The AAA bases must have changed into a valine codon
B) The anticodon can select the AAA position while incorrect charging supplies valine there
C) The ribosome necessarily converts the attached valine chemically into lysine
D) Amino-acid charging and codon recognition are the same molecular recognition event

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Q8 Report P describes needing increasing doses of an opioid to obtain a formerly achieved effect. Report Q describes withdrawal symptoms when repeated use is stopped. A third report identifies cocaine as a stimulant despite its plant origin. Which interpretation is correct?
A) P and Q both mean that a protective immune response has developed
B) P proves physical dependence, while Q establishes only a higher dose requirement
C) Plant origin makes cocaine and all opioids identical receptor agonists
D) P describes tolerance, Q supports dependence, and plant origin alone does not determine drug class or action

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Q9 A transgenic animal expresses a human therapeutic protein in milk, followed by product purification. Separately, a company uses a community's biological resource and documented traditional knowledge without proper authorisation or benefit sharing. Which distinction is appropriate?
A) Successful transgene expression by itself proves that every related patent claim is justified
B) Purification of a therapeutic protein is the definition of biopiracy
C) The first is a transgenic production application; the second raises biopiracy concerns, and expression alone does not settle provenance or ethical permission
D) Traditional knowledge can never be relevant to the use of a biological resource

Q10 Four biodiversity losses are observed: a forest is divided into isolated patches; an introduced predator eliminates native island prey; a commercially harvested species declines under excessive collection; and an obligately pollinated plant disappears after its sole pollinator is lost. Which sequence best identifies the main drivers?
A) Invasion, fragmentation, co-extinction, overexploitation
B) Habitat fragmentation, invasion, overexploitation, co-extinction
C) Overexploitation, co-extinction, invasion, fragmentation
D) Co-extinction, overexploitation, fragmentation, invasion

Q11 A large recipient population has 200 males and 200 females, with genotype frequencies AA=0.40,Aa=0.20,aa=0.40 separately in each sex. It receives 100 male immigrants with frequencies AA=0.64,Aa=0.32,aa=0.04. Each male contributes equally to the sperm pool and each female equally to the egg pool; mating is random between these pools. What are the pooled adult A-allele frequency and expected heterozygote frequency in the next generation?
A) 0.50 and 0.50
B) 0.80 and 0.32
C) 0.64 and 0.4608
D) 0.56 and 0.50

Q12 An origin-of-life model circulates a reducing gas mixture through water vapour and electrical discharges in a closed apparatus. Organic compounds, including amino acids, accumulate, but no self-maintaining reproducing cells are detected. What is the strongest warranted conclusion?
A) The apparatus has demonstrated the complete origin of a living cell
B) Every modern atmospheric composition must produce the same products
C) Abiotic synthesis of some organic building blocks is possible under the tested conditions; the origin of a living cell has not thereby been demonstrated
D) The organic products prove that electrical energy cannot drive chemical change

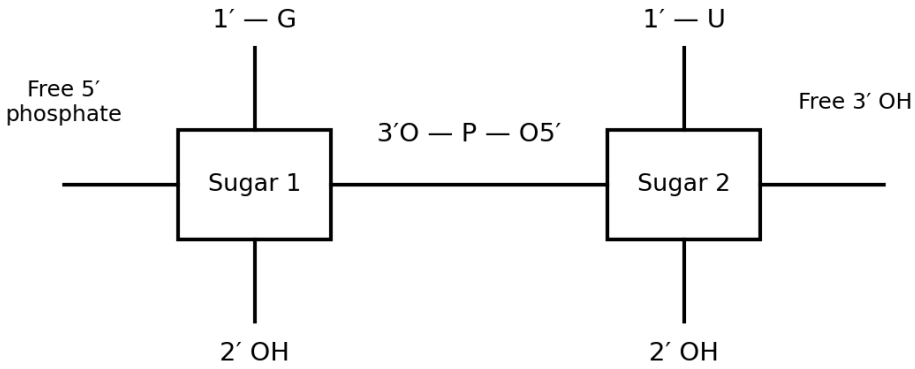
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Q13 During quiet inspiration, the diaphragm contracts and flattens and thoracic volume increases. Which pressure change draws air inward, and what normally happens during the following quiet expiration?
A) Alveolar pressure falls slightly below atmospheric pressure; relaxation reduces thoracic volume and raises alveolar pressure slightly above atmosphere
B) Alveolar pressure rises above atmosphere during inspiration; further diaphragm contraction drives expiration
C) Atmospheric pressure must fall to zero; expiration then needs no pressure difference
D) Both phases require alveolar pressure to remain exactly equal to atmosphere throughout airflow

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Q14 The nucleotide diagram shows a strand with a free 5-prime phosphate at the left, 3-prime-to-5-prime phosphodiester connections, and a free 3-prime OH at the right. The illustrated sugars have a 2-prime OH and one base is uracil. Which assignment and chain-extension site follow?

Nucleotide polarity: labelled connectivity



Boxes show sugar attachment positions; bases and OH labels are explicit.

A) DNA; add the next nucleotide to the left 5-prime phosphate
B) RNA; add the next nucleotide to a base’s amino group
C) RNA; add the next nucleotide to the right 3-prime OH
D) DNA; add the next nucleotide to the right 2-prime OH

Q15 Two engineered cultures express the same target protein per viable cell in a small aerated flask. On scale-up, vessel P has poor oxygen transfer and falling viable-cell density. Vessel Q maintains aeration but loses much target during separation and purification. Which distinction best explains why equal small-flask expression does not guarantee equal final product yield?
A) Oxygenation and downstream recovery are irrelevant once the gene is present
B) P has an upstream growth limitation, while Q has a downstream recovery limitation; expression is only one factor in final yield
C) Q must have lost every transgene because purification losses are DNA mutations
D) P and Q must have equal product yield because per-cell expression was once equal

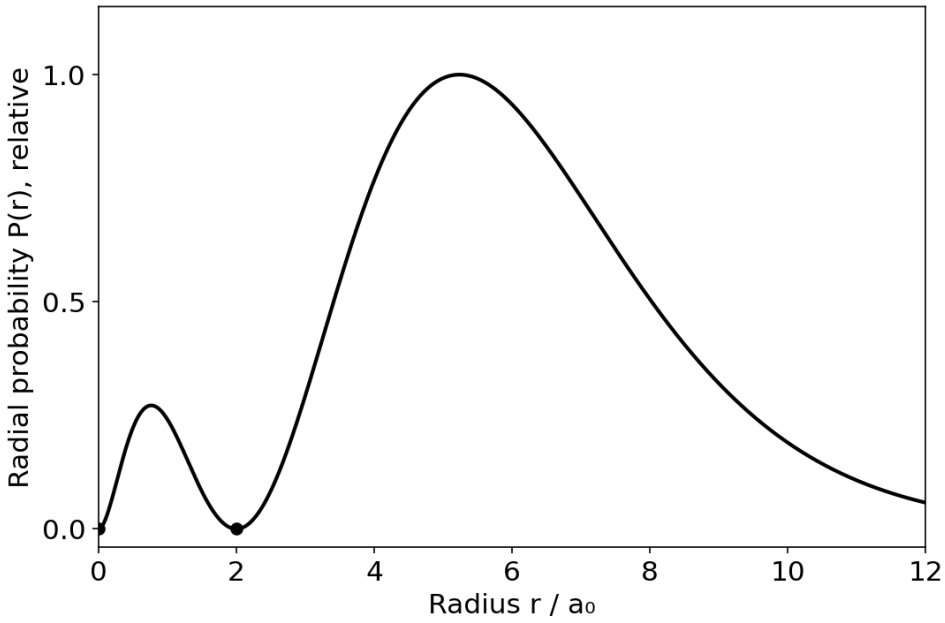
Chemistry

Q16 For CH2=CH–C≡C–CH2–CHO, count only the six carbon atoms and use the usual local hybridisation model. What are the numbers of sp, sp2 and sp3 carbons, respectively?
A) 2,3,1
B) 2,2,2
C) 1,4,1
D) 3,2,1

Q17 Compounds P=[Co(NH3)6][Cr(CN)6] and Q=[Cr(NH3)6][Co(CN)6] are considered with NH3 neutral, CN− monodentate and both metals in oxidation state +3. Which relationship and charge accounting are correct?
A) Linkage isomers; every complex ion is neutral
B) Coordination isomers; each has a 3+ complex cation and a 3− complex anion
C) Ionisation isomers; CN− changes from monodentate to bidentate
D) Geometrical isomers; one metal must change oxidation state

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Q18 The angular probability distribution of an orbital is spherical. Its radial probability P(r) is graphed: it vanishes at r=0 and at one positive radius, but is positive on both sides of that interior zero. Also P(r)/r2 approaches a positive constant at the origin. Use P(r) proportional to r2 times the local probability density, angular-node count l and radial-node count n-l-1. Which orbital and nuclear-position density follow?

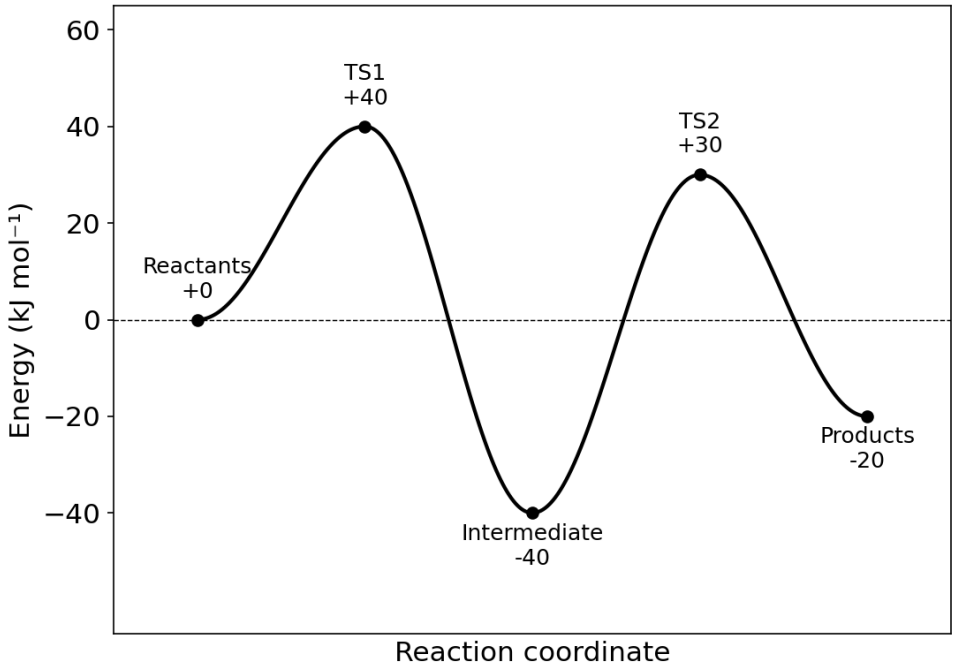


Angular distribution spherical; P(r)/r² has a positive limit at r=0.

A) 1s; the density must be zero because P(0)=0
B) 2p; spherical angular probability requires l=1
C) 2s; the local density at the nucleus is nonzero despite P(0)=0
D) 3s; every zero including the radial origin counts as a radial node

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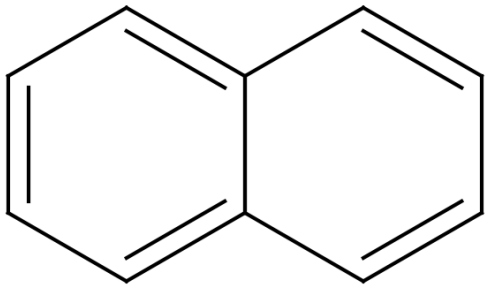
Q19 The two-step reaction profile has reactants at 0, first transition state at +40, intermediate at -40, second transition state at +30 and products at -20 kJ mol−1. What are the larger local forward activation barrier and the overall reaction enthalpy?



A) 40 and -20 kJ mol−1
B) 30 and -20 kJ mol−1
C) 70 and +20 kJ mol−1
D) 70 and -20 kJ mol−1

Q20 The figure shows a fused two-hexagon aromatic hydrocarbon with one shared edge; every unlabelled vertex is carbon and only the required hydrogens are omitted. What formula and interpretation of the alternating-bond drawing are correct?

Fused aromatic skeleton: one bond contributor



Unlabelled vertices are carbon; hydrogens omitted.

A) C12H12; the rings are separate cyclohexanes
B) C10H8; the drawing is one resonance contributor, not proof of permanently localised alternating bonds
C) C12H10; the rings share no carbon atoms
D) C10H10; every carbon has one hydrogen

Q21 Chlorobenzene undergoes electrophilic nitration more slowly than benzene under comparable conditions, yet its principal substitution positions are ortho and para. Which explanation reconciles rate and orientation?
A) Chlorine's inductive withdrawal can deactivate the ring while lone-pair resonance influences the relative stability of ortho/para attack pathways
B) Every electron-withdrawing substituent must direct exclusively meta, so the observations are impossible
C) A slower rate proves that chlorine cannot donate any electron density by resonance
D) Ortho/para direction proves that chlorobenzene must react faster than benzene

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Q22 Two Williamson routes are proposed for phenetole, Ph–O–CH2CH3, under ordinary SN2 conditions: P uses sodium phenoxide and bromoethane; Q uses sodium ethoxide and bromobenzene. Which assessment is correct?
A) Both routes are equivalent because their total atom counts match
B) Only Q is suitable because an aryl carbon is the easier SN2 centre
C) P is suitable; Q incorrectly treats the aryl C–Br centre as an ordinary SN2 substrate
D) Neither route can form an ether because alkoxides never act as nucleophiles

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Q23 Pentan-2-one and pentan-3-one are candidates for an unknown ketone. Under the specified oxidative-cleavage conditions both candidates can give one ethanoic-acid and one propanoic-acid fragment. The original carbonyl carbon is 13C-labelled, remains in one acid's carboxyl group, and no carbon scrambling occurs. The label is found in ethanoic acid. Which inference follows?
A) Only pentan-3-one fits because its carbonyl carbon has a methyl neighbour
B) Neither candidate can contain five carbon atoms
C) The unlabelled product pair already uniquely identifies pentan-2-one
D) Pentan-2-one fits; pentan-3-one would retain its carbonyl carbon with an ethyl group in labelled propanoic acid

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Q24 Aniline is brominated with excess bromine water to give the usual 2,4,6-tribromo derivative. Its amino group is then diazotised at 0–5 degrees C and replaced by hydrogen using H3PO2, with the bromines retained. Which final aromatic product is obtained after renumbering the ring?
A) 1,3,5-Tribromobenzene
B) 2,4,6-Tribromoaniline
C) 2,4,6-Tribromophenol
D) Bromobenzene

Q25 One mole of solvent is mixed ideally with 0.10 mol nonvolatile solute X and 0.15 mol nonvolatile solute Y. Neither associates nor dissociates. What is the final solvent vapour pressure divided by its pure vapour pressure, and why is it not 1/1.10?
A) 0.90; only X can affect vapour pressure
B) 0.80; both solutes enter the total mole count
C) 0.75; mole fraction equals one minus total solute moles
D) 1.25; adding solute increases solvent mole fraction

Q26 Standard reduction potentials are A3+ + e− → A2+, E1°=+0.50 V, and A2+ + 2e− → A(s), E2°=-0.10 V. What is E° for A3+ + 3e− → A(s), and its standard Gibbs energy change expressed using Faraday constant F?
A) +0.40 V; -1.20F
B) +0.20 V; -0.60F
C) +0.10 V; -0.30F
D) -0.10 V; +0.30F

Q27 A reaction has intrinsic rate law r=k[A][B]. In two runs B is maintained in large excess so its concentration changes negligibly. Run P uses [B]=0.20 M and run Q uses [B]=0.40 M. What logarithmic A-time behaviour and interpretation follow?
A) Neither run can have a straight ln[A] plot because the intrinsic overall order is two
B) Both show straight ln[A] plots with equal slope because the same temperature fixes k
C) Q must be zero order because it contains more B
D) Both can show apparent first-order A decay, with Q's slope magnitude twice P's, while intrinsic overall order remains two

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Q28 At the same temperature, neutralising 0.100 mol strong acid with strong base releases 5.70 kJ. Neutralising 0.100 mol weak monoprotic acid HA with the same base releases 5.20 kJ as measured. A matched dilution blank for the weak-acid run releases 0.20 kJ and is included in that measured heat. What molar enthalpy of HA ionisation follows? Assume the remaining reaction steps are identical.
A) +7.0 kJ mol−1
B) +5.0 kJ mol−1
C) -7.0 kJ mol−1
D) +0.70 kJ mol−1

Q29 For $CaCO_3(s) \rightleftharpoons CaO(s)+CO_2(g)$, $K_p=0.40$ bar at a fixed temperature. Two identical rigid vessels have $RT/V=1.00$ bar mol^{-1} and initially contain no CO_2 but 0.50 mol CaO. P contains 1.00 mol $CaCO_3$; Q contains only 0.10 mol $CaCO_3$. What are the final CO_2 pressures, allowing a solid phase to be exhausted?
A) P=0.40 bar, Q=0.40 bar
B) P=0.40 bar, Q=0.10 bar
C) P=1.00 bar, Q=0.10 bar
D) P=0.10 bar, Q=0.40 bar

Q30 Use the neutral configurations Ce: [Xe]4f1 5d1 6s2, Eu: [Xe]4f7 6s2 and U: [Rn]5f3 6d1 7s2. In the usual formal valence-electron accounting for Ce^{4+}, Eu^{2+} and U^{6+}, which f counts and interpretation are consistent?
A) Ce 4f1, Eu 4f5, U 5f3; oxidation removes only s electrons forever
B) Ce 4f0, Eu 4f7, U 5f0; these states arise because atomic mass alone fixes oxidation number
C) Ce 4f0, Eu 4f7, U 5f0; empty or half-filled arrangements can matter, and nearby actinoid valence-orbital energies permit wider oxidation-state variation
D) Ce 4f4, Eu 4f9, U 5f6; positive charge adds f electrons

Mathematics

Q31 Let $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, which satisfies A²-3A+2I=0. Find det$(A^5 - 3A^4 + 4A^2 - I)$.

A) 1
B) -1
C) 3
D) -3

Q32 Five tiles consist of two identical tiles marked AB, two identical tiles marked C and one tile marked D. Tiles remain intact in a row. How many distinct rows have D at either end?

A) 6
B) 24
C) 30
D) 12

Q33 Let f:{1,2,3}→{a,b,c,d} map 1,2,3 to a,b,c respectively. How many maps g:{a,b,c,d}→{0,1} make g∘f onto {0,1}, and what does this require of f and g?

A) 12; g must be onto, while f need not be onto its stated codomain
B) 12; f must be onto, while g need not be onto
C) 14; both factors must be onto
D) 6; neither factor can be onto

Q34 A real-coefficient cubic is x³+ax²+bx-10 and has root 1+2i. What is the sum of the squares of all three roots, counting multiplicity?

A) 2
B) 10
C) -2
D) 14

Q35 Evaluate $\int_{-2}^{2} \left|\min\{x^2, 2 - x^2\}\right| \, dx$.

A) 0
B) 8(√2-1)/3
C) 16√2/3
D) 16(√2-1)/3

Q36 Lines L1=(1,0,2)+t(1,2,-1) and L2=(b,2,1)+s(2,k,-2) depend on real b,k. Which pair makes them the same line, rather than merely parallel?

A) k=4,b=4
B) k=4,b=2
C) k=2,b=4
D) k=-4,b=2

Q37 For the real curve y²=x⁴, the formula y'=2x³/y gives 0/0 at the origin. What are the tangent lines there when both real branches are considered?

A) Both branches have the horizontal tangent y=0
B) Both branches have the vertical tangent x=0
C) The two branches have slopes +1 and -1
D) No real branch passes through the origin

Q38 A data set has n observations, mean 2 and population variance 3. A new set of 2n observations contains every original x once and its reflected value 8-x once. What are the new mean and population variance?
A) 4 and 3
B) 2 and 3
C) 4 and 7
D) 6 and 7

Q39 Let ⌊x⌋ be the greatest integer not exceeding x. Find $\lim_{x\to+\infty} x \sin\frac{1}{\lfloor x\rfloor}$.
A) 0
B) Does not exist
C) 2
D) 1

Q40 Box A contains one red and one blue ball; box B contains three red and one blue. In protocol P, choose a box with probability 1/2 each and then choose a ball uniformly from it. In protocol Q, choose uniformly from all six balls while retaining each ball's box label. Given a red ball, what are P(B|red) in protocols P and Q, respectively?
A) 3/5 and 3/4
B) 3/4 and 3/5
C) 1/2 and 1/2
D) 3/5 and 1/2

Q41 For 0≤x≤5π/2, consider sin(2x)cos x>0. How many connected solution intervals are there, and what is their total length?
A) Two intervals; 3π/2
B) Three intervals; 3π/2
C) Three intervals; 5π/2
D) Two intervals; π

Q42 The solution of $y'+2y/(x-1) = 1/(x-1)^2$, y(0)=1, is taken on its maximal real interval containing zero. What are that interval and the minimum value of y there?
A) (-∞,∞); -1/8
B) (-∞,1); 0
C) (-∞,1); -1/8
D) (1,∞); -1/8

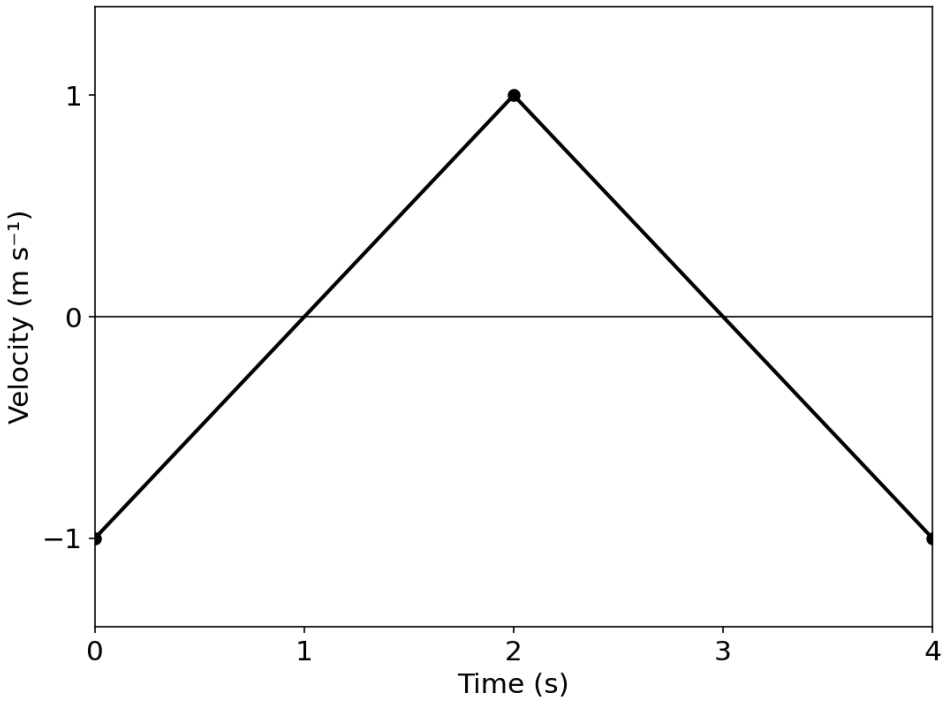
Q43 For which real a does $x^2-(a+1)x+2a-1=0$ have real roots, not necessarily distinct, whose product is positive?
A) (1/2,1]∪[5,∞)
B) [1,5]
C) (1/2,5)
D) (-∞,1]∪[5,∞)

Q44 Define rotated coordinates u=(3x+4y)/5 and v=(-4x+3y)/5. Which equation describes the line 3x+4y=10 in these coordinates, and what is its perpendicular distance from the origin?
A) v=2; distance 2
B) u=10; distance 10
C) u=2; distance 10
D) u=2; distance 2

Q45 A plane is oriented by the ordered points A=(1,0,0), B=(0,1,0), C=(0,0,1), with unit normal along AB×AC. For P=(1,1,1), which normal and signed perpendicular displacement from the plane are correct?
A) n=(1,1,1)/√3; displacement -2/√3
B) n=(-1,-1,-1)/√3; displacement 2/√3
C) n=(1,1,1)/√3; displacement 2/√3
D) n=(1,1,1); displacement 2

Physics

Q46 A particle’s velocity graph is v=t-1 m s−1 for 0≤t≤2 s and v=3-t m s−1 for 2≤t≤4 s, with t entered in seconds. What are its displacement and total distance over these four seconds?



A) 0 m and 0 m
B) 0 m and 2 m
C) 2 m and 2 m
D) 0 m and 4 m

Q47 A mass m on a light inextensible rope of length L moves in a vertical circle under gravity alone apart from rope tension. At a point 60 degrees from the downward vertical its speed is v. The rope breaks if tension exceeds 8mg. What is the greatest v compatible with completing the circle without breaking the rope anywhere?
A) √(15gL/2)
B) √(8gL)
C) √(6gL)
D) √(4gL)

Q48 A body is moved quasistatically from A to B along one rough path with external work +10 J, then from B to A along another rough path with external work -4 J. It returns to the same mechanical state, all friction is dissipative, and the only stored-energy change on an individual leg is a conservative potential U. What are total dissipated heat and the possible range of UB-UA?
A) 6 J; 4≤UB-UA≤10 J
B) 14 J; UB-UA=10 J
C) 6 J; UB-UA=6 J
D) 0 J; 4≤UB-UA≤10 J

Q49 A uniform solid cylinder of mass M and radius R rolls without slipping up a 30-degree incline. A pure external torque MgR is applied in the rotation direction for uphill rolling and exerts no net force. Use ICM=MR²/2 and assume enough static friction. What are its centre acceleration and friction force?
A) a=g uphill; friction Mg/2 downhill
B) a=g/3 uphill; friction Mg/6 downhill
C) a=g/2 uphill; friction Mg uphill
D) a=g/3 uphill; friction 5Mg/6 uphill

Q50 A satellite initially has circular speed vc=√(GM/r) at radius r. A short impulse adds a velocity increment either radially outward or in the original tangential direction. Neglect displacement during the impulse. What are the minimum increment magnitudes for escape in these two cases, respectively?
A) (√2-1)vc and vc
B) vc and (√2-1)vc
C) √2 vc and √2 vc
D) vc/2 and vc/2

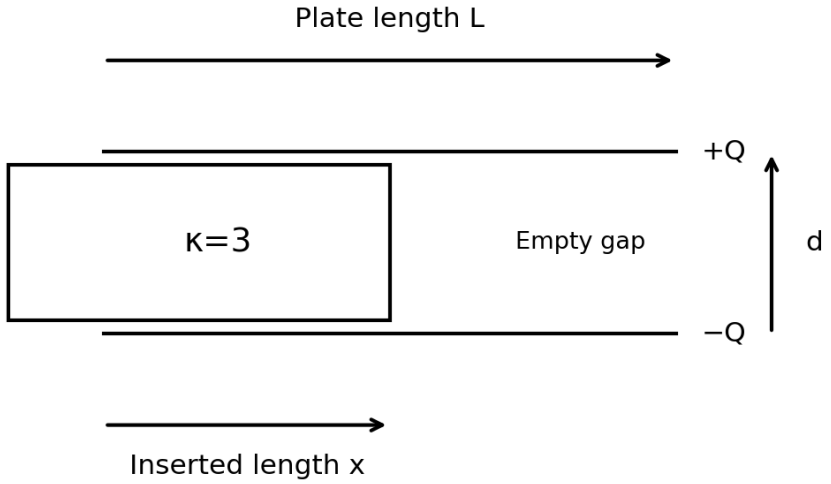
Q51 A soap film spans a frame with a movable side of length 0.10 m. It is extended quasistatically by 0.020 m at constant surface tension 0.030 N m−1. Ignore gravity and other forces. What is the work and its ratio to making the same area increase at one liquid–air interface?
A) 6.0×10−5 J; ratio 1
B) 1.2×10−4 J; ratio 1
C) 1.2×10−4 J; ratio 2
D) 2.4×10−4 J; ratio 2

Q52 One mole of monatomic ideal gas is compressed quasistatically along PV^(3/2)=constant from V0 to V0/4, starting at temperature T0. Use Cv=3R/2 and the convention ΔU=Q-Wby. What heat Q enters the gas?
A) -RT0/2
B) +RT0/2
C) +3RT0/2
D) 0

Q53 Two spheres of radius R each carry total positive charge Q. One is a nonconducting sphere with uniform volume charge; the other is a conductor in electrostatic equilibrium. A small positive test charge q is moved quasistatically from the surface to the centre without changing the distributions. What work is done by the electric field in the two cases, respectively?
A) +kQq/(2R) and 0
B) 0 and -kQq/(2R)
C) -kQq/R and -kQq/R
D) -kQq/(2R) and 0

Q54 An isolated parallel-plate capacitor initially has capacitance C0=ε0 bL/d and fixed charge Q. A dielectric of relative permittivity 3 fills the full gap over width b and inserted length x, leaving the remaining plate length L-x empty. Use the parallel-area capacitance approximation, neglecting fringing corrections to C. What is the dielectric's force magnitude at x=L/2, and its direction?

Isolated capacitor with partial dielectric overlap

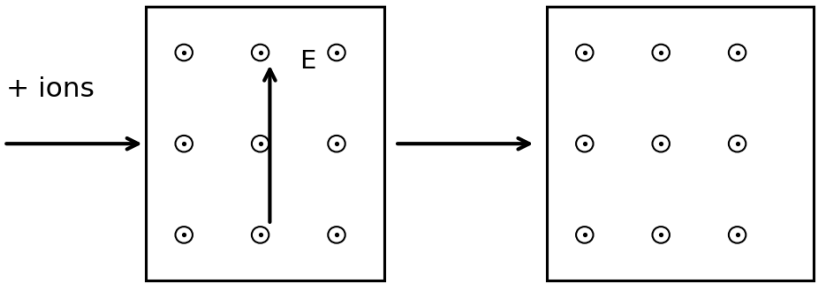


Width b is perpendicular to page; no battery.

A) $Q^2/(C0L)$; outward
B) $Q^2/(4C0L)$; inward
C) $Q^2/(2C0L)$; inward
D) 0; no battery is attached

Q55 Two positive ion beams have masses and charges (m,q) and (4m,2q). They pass undeflected through the same crossed-field velocity selector, then enter the same uniform magnetic analyser perpendicular to its field. What are the second beam's radius divided by the first's, and its kinetic energy divided by the first's?

Crossed-field selector followed by magnetic analyser

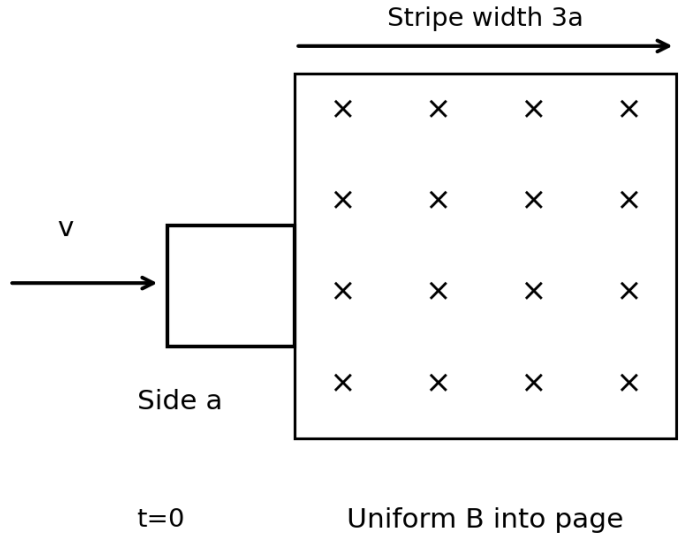


Selected beams enter the analyser at the same velocity.

A) 1; 2
B) √2; 2
C) 2; 4
D) 4; 4

Q56 A square conducting loop of side a moves right at speed v through a stationary field stripe of width 3a, whose magnetic field points into the page. At t=0 its leading edge reaches the stripe's left boundary. Neglect self-inductance. During which open time interval is the induced current zero while the loop is wholly within the field?

Moving loop at its first contact with a field stripe



A) 0<t<a/v
B) a/v<t<3a/v
C) 3a/v<t<4a/v
D) It is never zero because the loop always moves

Q57 A right-angled isosceles glass prism has refractive index √3. A ray enters normally through a short face and reaches the hypotenuse at internal incidence 45 degrees. Compare the exit at that face when the surrounding medium is air and when the whole prism is immersed in transparent liquid of index 4/3. Which result follows?
A) Air gives total internal reflection; liquid permits emergence with sin r=3√6/8
B) Both media give total internal reflection because the prism angle is unchanged
C) Both media permit emergence with sin r=√6/2
D) Air gives total internal reflection; liquid permits emergence with sin r=√6/2

Q58 Two equal coherent slit contributions have individual intensity I0. In P their phase is stable. In Q an extra phase alternates rapidly between 0 and π for equal times, and the detector averages over many switches. What are Q's mean intensities at P's bright and dark positions, and how would a constant extra π phase differ?
A) 4I0 and 0; constant π would erase all fringes
B) 0 and 4I0; constant π gives the same uniform result
C) I0 and I0; constant π doubles the wavelength
D) 2I0 and 2I0; constant π would interchange bright and dark positions rather than wash out contrast

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Q59 A hydrogen-like one-electron ion has orbit radius 2a0 and electron de Broglie wavelength πa0, where a0 is the hydrogen ground-state Bohr radius. Use $2\pi r = n\lambda$, $r = a_0 n^2/Z$ and $E_n = -13.6\ Z^2/n^2$ eV. Which n,Z and energy fit all three conditions?
A) n=2,Z=2,E=-13.6 eV
B) n=4,Z=4,E=-13.6 eV
C) n=8,Z=32,E=-217.6 eV
D) n=4,Z=8,E=-54.4 eV

Q60 A radioactive nuclide obeys exponential decay with half-life T. One nucleus has already survived for 3T; another nucleus of the same nuclide has just been produced. Their environments are identical. What are their respective probabilities of surviving the next interval T, and can their past histories fix which one decays first?
A) 1/2 and 1/2; neither past history fixes the order
B) 1/16 and 1/2; the older survivor must decay first
C) 1/2 and 1; the newly produced nucleus cannot decay during its first half-life
D) 0 and 1/2; no nucleus can survive more than three half-lives

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / D Diagnostic body structures distinguish five higher nonchordate groups without relying on names alone

Answer: (D) Arthropoda, Mollusca, Echinodermata and Hemichordata.

Why: Jointed appendages and an exoskeleton identify the arthropod combination. A mantle and muscular foot distinguish the molluscan plan. The water-vascular system, considered with adult pentaradial organisation, supports Echinodermata. The three-region body of proboscis, collar and trunk is characteristic of hemichordates; it must not be replaced by a chordate identity merely because of the historical name. Each assignment uses a distinctive structure rather than a general feature such as having a body cavity.

Why not the others: (A) exchanges arthropod and annelid characters and misassigns the last two groups. (B) gives a mantle-bearing animal an annelid identity and misclassifies S. (C) ignores the echinoderm water-vascular system.

Key point: Use diagnostic combinations to navigate a classification key.

Q2 / B A selective wall defect separates collenchymatous support from lignified fibres and vessels

Answer: (B) Collenchyma and lignified support/conducting elements depend on different wall features.

Why: P's living cells and uneven corner thickening identify collenchyma, whose support does not depend on the same heavily lignified wall pattern as sclerenchymatous Q. R's perforated tube is a xylem vessel element arrangement; lignified walls help it withstand mechanical stresses and resist collapse during water transport. Being dead at maturity does not make Q and R independent of the wall properties established during differentiation. The specified defect therefore affects the tissues differently.

Why not the others: (A) assumes an immediate cell-identity conversion not caused by the stated defect. (C) destroys the explicitly preserved thickening mechanism and ignores lignified walls in dead mature cells. (D) treats loss of one wall component as automatic conversion between vascular tissues.

Key point: Link wall composition to function without replacing tissue identity.

Q3 / A Meiotic pairing and exchange ordered separately from the two segregation events

Answer: (A) Pairing precedes crossing over, followed by the two different segregation events.

Why: Homologous pairing develops during zygotene, and crossing over between nonsister chromatids occurs during pachytene of prophase I. At anaphase I the homologous chromosomes move apart without the sister centromeres dividing in the ordinary meiotic pattern. Sister chromatids are separated during the second meiotic division. The chromosome relationships therefore differ at the two anaphases even though both move genetic material toward opposite poles.

Why not the others: (B) reverses the pairing/exchange sequence and assigns the wrong segregation event to division I. (C) eliminates the defining distinction between the two divisions. (D) assigns homologous synapsis and crossing over to ordinary mitosis.

Key point: Track homologues and sisters as different chromosome relationships.

Q4 / C A blocked return intermediate distinguishes transient C4 decarboxylation from sustained acceptor regeneration

Answer: (C) Existing intermediates can support a transient step, but the carbon shuttle cannot be sustained.

Why: The initial carbon acceptor in the mesophyll is PEP. Its regeneration uses the returning three-carbon intermediate and requires energy. Existing four-carbon acid can initially reach a functioning decarboxylation step, so a transport block need not instantly poison RuBisCO or erase every intermediate. Over time, failure to return and regenerate the acceptor restricts continued operation of the concentrating pathway. The Calvin cycle remains associated with bundle-sheath CO2 fixation in the stated C4 arrangement.

Why not the others: (A) invents relocation of the Calvin cycle as an automatic response. (B) confuses a metabolic intermediate with a protein. (D) reverses the energetic requirement of PEP regeneration.

Key point: Separate a transient use of existing intermediates from a pathway's sustained regeneration requirement.

Q5 / D Preserved thyroxine feedback does not exclude loss of parathyroid calcium regulation

Answer: (D) PTH and calcitonin act in calcium regulation, whereas normal thyroxine does not replace PTH.

Why: Parathyroid hormone is a calcium-raising hormone, and its loss can account for the stipulated low-calcium effect in this model. Thyroid calcitonin has an opposing influence on blood calcium, but it is not the same substance as thyroid hormone involved in metabolic regulation. Normal thyroxine and its feedback therefore cannot be used as a test of intact parathyroid output. The question fixes the lesion and asks for hormonal interpretation rather than a unique diagnosis from symptoms alone.

Why not the others: (A) treats anatomically nearby endocrine sources as functionally identical. (B) reverses the calcium effect of PTH and changes the stipulated lesion. (C) assigns direct calcium supply to a pituitary signal.

Key point: Identify each hormone's source and regulated variable separately.

Q6 / A Three stage-and-site constraints distinguish later embryo transfer, GIFT and ICSI

Answer: (A) Further culture with intrauterine transfer, GIFT and ICSI match the three constraints.

Step 1: Early embryos up to eight blastomeres can normally be transferred to the tube, but P's blocked tubes invalidate that route. Further culture permits transfer of a later embryo to the functional uterus.

Step 2: Q's aim is fertilisation within her body using a donor ovum. Gamete intrafallopian transfer provides the appropriate unfertilised gamete and patent-tube site.

Step 3: Direct sperm injection into the oocyte is intracytoplasmic sperm injection, a different operation from placing semen in the reproductive tract.

Why: Embryonic stage, anatomical access and whether fertilisation has already happened all constrain the route.

Why not the others: (B) ignores the tubal obstruction. (C) substitutes an embryo for the required unfertilised ovum. (D) confuses an intrauterine insemination site with the oocyte cytoplasm.

Key point: Match reproductive technology to stage, site and biological purpose.

Q7 / B A deliberately mischarged adaptor separates amino-acid specificity from codon decoding

Answer: (B) Codon placement and amino-acid attachment are distinct specificity steps.

Why: The anticodon pairs antiparallel with the AAA codon, directing where this tRNA is used. The amino acid delivered is the molecule actually attached to the adaptor, here stipulated to be valine after editing has been bypassed. Normally aminoacyl-tRNA synthetase recognition and associated checks help maintain the correct amino-acid assignment. This experiment separates that charging specificity from the codon–anticodon interaction rather than changing the genetic code or the mRNA sequence.

Why not the others: (A) invents an RNA sequence mutation. (C) assigns amino-acid conversion to the ribosome without any such reaction. (D) ignores that different sites and recognition processes determine charging and decoding.

Key point: An adaptor's anticodon and its attached amino acid convey different information.

Q8 / D Dose-response adaptation separated from withdrawal and plant-source classification

Answer: (D) Tolerance, dependence and pharmacological class must be distinguished.

Why: Tolerance refers to reduced response to a previously effective dose, so the required dose may rise. Dependence is supported by a withdrawal syndrome after discontinuation in the stated history. These phenomena can coexist but are not interchangeable definitions. Drug source is another independent property: cocaine's plant origin does not make its stimulant action or molecular targets identical to opioid actions. The reports therefore describe different dimensions of drug use.

Why not the others: (A) confuses nervous-system adaptation with immunity to infection. (B) exchanges the two operational observations. (C) classifies receptor action from source alone and contradicts the stimulant description.

Key point: Classify the measured effect rather than treating every repeated-use change as dependence.

Q9 / C Transgenic protein production separated from resource-provenance and biopiracy concerns

Answer: (C) Technical production and the ethical provenance of resource use are different questions.

Why: Transgenic animals can be used to produce biologically useful human proteins; recovery and purification are then required to obtain a usable product. Biopiracy concerns unauthorised or unfair commercial use of biological resources and associated knowledge, as stipulated in the separate case. Demonstrating that a gene can be expressed does not establish the origin of the resource, consent or fair sharing. The question distinguishes biological application from the ethical issue rather than deciding the validity of a particular real patent.

Why not the others: (A) substitutes technical success for an assessment of provenance and permission. (B) misdefines a processing step as resource appropriation. (D) dismisses the very knowledge specified as part of the resource use.

Key point: Separate what a technology does from how its inputs and knowledge were obtained.

Q10 / B Four biodiversity-loss drivers identified by their distinct causal observations

Answer: (B) The four observations correspond to fragmentation, invasion, overexploitation and co-extinction.

Why: Dividing habitat reduces continuity and can isolate populations even where some habitat remains. An introduced predator adds an invasive biotic pressure. Excessive removal directly exceeds the harvested population's capacity to persist. In the final case, loss of an indispensable interacting species causes a dependent species to disappear, which is co-extinction. These drivers can interact in real ecosystems, but the observations specify the principal causal change for each case.

Why not the others: (A) exchanges the first two and last two mechanisms. (C) treats collection and introduction as though they were the same process. (D) attributes dependence-related loss to a case where habitat division is the specified cause and misassigns the other observations.

Key point: Identify the causal change, not only the shared outcome of population loss.

Q11 / D Male-only gene flow separates adult allele pooling from sex-specific gamete contributions

Answer: (D) Adult pooled A frequency is 0.56, while offspring heterozygosity is 0.50.

Given: Migration is male-only and gametic contributions are equal within each sex, not across all adults regardless of sex.

Formula/Principle: Count adult alleles separately from the sperm and egg frequencies used to form offspring.

Step 1: Resident and immigrant A frequencies are

$$p_R = 0.40 + 0.20/2 = 0.50, \quad p_I = 0.64 + 0.32/2 = 0.80.$$

Adult pooling gives

$$p_{adult} = \frac{400(0.50) + 100(0.80)}{500} = 0.56.$$

Step 2: Sperm frequency is

$$p_s = \frac{200(0.50) + 100(0.80)}{300} = 0.60,$$

while egg frequency remains pe=0.50.

Step 3: Heterozygotes arise from A sperm with a egg or a sperm with A egg.

$$f_{Aa} = p_s(1 - p_e) + (1 - p_s)p_e = 0.60(0.50) + 0.40(0.50) = 0.50.$$

Why: Male-only immigration makes the two parental allele pools different. Applying 2p(1-p) to the pooled adult p would ignore the stated sex-specific gametic contributions. No founder bottleneck is required.

Why not the others: (A) ignores the immigrant effect on the adult allele count. (B) uses only immigrant frequencies. (C) treats the immigrant AA genotype frequency as an allele-pool quantity and applies the wrong mating model.

Key point: Construct the gamete pools actually used in mating before predicting offspring frequencies.

Q12 / C Abiotic organic synthesis distinguished from demonstrated cellular life

Answer: (C) The result demonstrates a chemical-synthesis possibility within its specified conditions.

Why: The model tests whether supplied nonliving starting materials and an energy source can yield organic molecules. Detecting amino acids supports that stage of chemical evolution. A self-maintaining, reproducing cellular system would require additional organisation and processes that the observation explicitly does not establish. The result also depends on the chosen gas mixture and apparatus, so it cannot be transferred without qualification to every atmospheric composition.

Why not the others: (A) equates building blocks with an autonomous living system. (B) ignores the experimental conditions as part of the conclusion. (D) contradicts organic synthesis occurring with electrical discharges.

Key point: Match a claim about life's origin to the level of organisation actually produced.

Q13 / A Quiet breathing linked through diaphragm motion, thoracic volume and pressure gradient

Answer: (A) A changing alveolar–atmospheric pressure difference directs airflow.

Why: Expansion of the thorax and lungs lowers alveolar pressure slightly below atmospheric pressure, allowing air to enter. During ordinary quiet expiration, relaxation and elastic recoil reduce lung volume and raise alveolar pressure slightly above atmosphere, driving air out. The diaphragm's movement changes the volume; it does not directly pull individual air molecules through the airways. At the transition between phases pressures can equalise, but sustained flow requires a pressure gradient.

Why not the others: (B) reverses the inspiratory gradient and the normal diaphragm action in quiet expiration. (C) invents an extreme external pressure change. (D) eliminates the driving gradient while asserting flow.

Key point: Connect muscle action to volume, then pressure, then airflow.

Q14 / C RNA sugar identity and phosphodiester polarity jointly locate the chain-extension site

Answer: (C) The structure is RNA, and normal chain extension uses the right-hand 3-prime OH.

Why: Ribose has the 2-prime hydroxyl shown, in contrast to deoxyribose's hydrogen at that position. Uracil provides a second compatible RNA feature. The backbone links a sugar's 3-prime position through phosphate to the next sugar's 5-prime position. A new nucleotide is therefore joined at the exposed 3-prime OH, extending the strand in the 5-prime-to-3-prime direction. Base amino groups do not provide the normal backbone extension site.

Why not the others: (A) contradicts the sugar and base evidence and extends the wrong end. (B) has the right nucleic-acid identity but an incorrect bonding site. (D) misidentifies the sugar and confuses the diagnostic 2-prime OH with the elongation 3-prime OH.

Key point: Use sugar composition to identify the strand and backbone polarity to locate its growth end.

Q15 / B Upstream oxygenation and downstream recovery separately limit scaled recombinant-product yield

Answer: (B) Upstream culture performance and downstream recovery can independently limit final yield.

Why: Product accumulated in a bioreactor depends on viable producing biomass and sustained physiological conditions, including adequate oxygenation for the stated culture. P's scale-up problem reduces that production capacity. Q instead loses material after production during processing, separation or purification. A per-cell expression measurement made in a small flask therefore omits both the number of productive cells over time and the fraction of their product recovered in usable form.

Why not the others: (A) treats gene presence as sufficient for the entire production process. (C) confuses physical recovery loss with a genetic change. (D) transfers a controlled small-scale measurement to different large-scale and processing conditions.

Key point: Distinguish expression, culture productivity and recoverable purified product.

Chemistry

Q16 / A Local hybridisation counts across alkene, alkyne, methylene and aldehyde centres

Answer: (A) Two sp, three sp2 and one sp3 carbon.

Step 1: Each of the two alkyne carbons has two electron-domain directions and is sp. Each of the two alkene carbons has three directions and is sp2.

Step 2: The aldehyde carbon also has three directions because its double-bonded oxygen counts as one direction, so it is sp2. The intervening CH2 has four sigma-bond directions and is sp3.

Why: Hybridisation is assigned locally; the presence of several multiple bonds does not make every carbon in the molecule the same type. Conjugation does not change the intervening saturated CH2 into a carbon with three bond directions.

Why not the others: (B) counts the carbonyl carbon as sp3. (C) counts only one of the two alkyne carbons as sp. (D) assigns an extra sp centre despite its three-direction bonding.

Key point: Count electron-domain directions separately at each carbon.

Q17 / B Ligand-set exchange between complex cation and anion preserves oxidation states and total composition

Answer: (B) They are coordination isomers with complex-ion charges +3 and -3.

Step 1: A six-ammine M(III) ion has charge +3. A six-cyanido M(III) ion has charge $+3 - 6 = -3$. Both written salts therefore balance electrically.

Step 2: The ligand sets are exchanged between two different metal centres in the complex cation and complex anion while the total composition is retained. This is coordination isomerism.

Why: Classifying the isomerism requires identifying what changes: the distribution of ligands between complex ions, rather than the donor atom of one ambidentate ligand or spatial arrangement within a fixed complex.

Why not the others: (A) invents a changed donor atom and miscounts ionic charges. (C) invokes a simple counterion exchange and a denticity change that do not occur. (D) changes neither metal's stated oxidation state and misidentifies the structural rearrangement.

Key point: Establish ionic composition before naming a coordination-isomer relationship.

Q18 / C An interior radial node and shell-volume factor distinguish 2s density from radial probability

Answer: (C) It is 2s, and the local probability density at the nucleus is nonzero.

Step 1: Spherical angular distribution identifies an s orbital, with l=0 and no angular nodes.

Step 2: The graph has one node at a positive radius. The radial origin is a boundary, not an additional nodal shell. Thus $n - l - 1 = 1$ gives n=2.

Step 3: The given relation means local density is proportional to P(r)/r². Its specified positive limit therefore gives nonzero density at the nucleus.

Why: Radial probability includes the size of a spherical shell. That geometrical factor vanishes at zero radius even when the local density does not. The graph must be interpreted as the stated radial quantity, not as the wavefunction itself.

Why not the others: (A) omits the interior node and confuses the two probability quantities. (B) contradicts the spherical angular distribution. (D) counts a boundary as an extra radial node.

Key point: Distinguish radial probability, local density and an interior nodal shell.

Q19 / D Two-step profile distinguishes local activation height from an absolute peak ordinate

Answer: (D) The larger local barrier is 70 kJ mol⁻¹ and the overall enthalpy is -20 kJ mol⁻¹.

Given: Each activation barrier is measured from the minimum immediately preceding its transition state.

Formula/Principle: Local barrier is transition-state energy minus its starting-state energy; overall enthalpy is final minus initial energy in this supplied profile.

Step 1: First-step barrier is $40 - 0 = 40$ kJ mol⁻¹. Second-step barrier is

$$30 - (-40) = 70 \text{ kJ mol}^{-1}.$$

Step 2: Overall change is

$$-20 - 0 = -20 \text{ kJ mol}^{-1}.$$

Why: The lower absolute second peak still rises farther above its own deeper intermediate. Comparing peak heights alone does not compare local activation energies. This calculation does not by itself assert an overall rate-limiting step without population assumptions.

Why not the others: (A) reports the first barrier. (B) uses an absolute peak ordinate as a barrier. (C) reverses the sign of the reactant-to-product change.

Key point: Measure each barrier from its own preceding minimum.

Q20 / B Fused aromatic vertex counting separated from a single localised-bond picture

Answer: (B) The fused aromatic structure is C10H8 and is described by resonance.

Step 1: Two six-carbon rings share the two carbons on their common edge, giving $6 + 6 - 2 = 10$ carbon atoms.

Step 2: The two shared junction carbons already bond to three other carbons and have no hydrogen. The remaining eight peripheral carbons each carry one hydrogen.

Why: A Kekulé-style alternating-bond sketch is a way of representing a conjugated aromatic system; it is not evidence that one fixed set of separate double bonds fully describes its electronic structure. Counting shared vertices also distinguishes fused rings from two rings linked by a single bond.

Why not the others: (A) ignores both fusion and unsaturation. (C) counts the two shared carbons twice and describes a different topology. (D) assigns hydrogens to the ring-junction carbons despite their valence.

Key point: Count the fused graph before adding hydrogens.

Q21 / A Halogen deactivation and ortho/para direction describe different kinetic comparisons

Answer: (A) Overall reactivity and relative positional preference need not have the same trend.

Why: The halogen's electron-withdrawing inductive effect lowers the ring's overall susceptibility to electrophilic substitution. Its lone-pair interaction can nevertheless influence the relative stabilisation of intermediates for attack at different positions, favouring ortho and para relative to meta. Deactivation compares the substituted ring with benzene, whereas orientation compares competing positions within that substituted ring. These are different comparisons, so the stated combination is consistent.

Why not the others: (B) applies an exceptionless rule that fails for halogens. (C) treats one net kinetic observation as proof that a competing electronic effect is absent. (D) equates positional direction with overall activation.

Key point: Keep the reference comparison for rate distinct from the comparison among substitution sites.

Q22 / C A Williamson target has one suitable reagent orientation and one invalid aryl-SN2 reversal

Answer: (C) Phenoxide attacks the primary alkyl halide in P; Q is not the equivalent reversed route.

Why: Williamson synthesis requires a suitable substitution centre. Bromoethane offers a primary sp3 carbon at which phenoxide can displace bromide. Reversing the partners leaves bromine on an aromatic sp2 carbon; ordinary backside SN2 substitution at that aryl centre is not the corresponding viable process. Atom balance is necessary for a synthetic proposal but does not establish the required mechanism.

Why not the others: (A) ignores which atom bears the leaving group. (B) reverses the usual substrate suitability. (D) denies the nucleophilic role of alkoxide or phenoxide oxygen.

Key point: Choose which side supplies the nucleophile and which supplies the substitutable alkyl carbon.

Q23 / D Carbonyl-carbon isotope retention resolves an otherwise ambiguous oxidative acid-fragment pair

Answer: (D) The isotope location, rather than the unlabelled acid pair alone, selects pentan-2-one.

Step 1: Pentan-2-one has a carbonyl carbon next to methyl and propyl groups. In the specified two-plus-three-carbon cleavage, retention with methyl gives labelled ethanoic acid.

Step 2: Pentan-3-one has an ethyl group on each side of its carbonyl. Retaining either side with that carbon gives a three-carbon labelled acid, propanoic acid.

Why: The supplied label tracks the original carbonyl carbon through a carbon-conserving cleavage. Without that tracking, the same pair of unlabelled fragment formulas would not distinguish the candidates under the stated conditions. No universal regioselectivity beyond the supplied cleavage model is assumed.

Why not the others: (A) gives pentan-3-one a methyl neighbour it does not possess. (B) ignores the conserved total of two plus three carbons. (C) discards the stated ambiguity that the isotope experiment resolves.

Key point: Track atom identity when product formulas alone are ambiguous.

Q24 / A A temporary amino directing group establishes an alternating bromine pattern before reductive removal

Answer: (A) 1,3,5-Tribromobenzene.

Step 1: The amino group activates the original ortho and para positions, giving bromines at positions 2,4,6 relative to the original NH2 carbon.

Step 2: Cold diazotisation converts NH2 to the diazonium group. H3PO2 replaces that group by H rather than by OH or CN. The three bromine positions are unchanged.

Step 3: Without the original amino substituent, renumbering from a brominated carbon gives the alternating pattern 1,3,5.

Why: A temporary directing group can establish a substitution pattern and then be removed; replacing the group does not move the other substituents.

Why not the others: (B) stops before diazotisation and deamination. (C) uses the hydrolysis product associated with water rather than the supplied reducing reagent. (D) removes two bromines without a stipulated reaction.

Key point: Preserve ring connectivity while renumbering after removal of a directing group.

Q25 / B A second nonvolatile solute changes the full solvent mole-fraction denominator

Answer: (B) The ratio is 0.80 because both nonvolatile solutes contribute particles.

Given: Solvent amount is 1.00 mol and total solute amount is 0.25 mol.

Formula/Principle: For an ideal solution with nonvolatile solutes, $p_{solvent} = x_{solvent}p^0$.

Step 1:

$$x_{solvent} = 1.00/(1.00 + 0.10 + 0.15) = 1/1.25 = 0.80.$$

Step 2: The ratio 1/1.10 would omit Y from the denominator even though it is another dissolved particle population.

Why: Raoult's law uses mole fraction in the entire liquid mixture, not a separately normalised solvent–X subsystem.

Why not the others: (A) neglects Y and also approximates the remaining ratio incorrectly. (C) subtracts amounts without dividing by total moles. (D) inverts the mole-fraction ratio.

Key point: Include every dissolved species in the mole-fraction denominator.

Q26 / C Electron-weighted Gibbs-energy addition combines unequal-electron reduction steps

Answer: (C) The combined potential is +0.10 V and ΔG°=-0.30F.

Given: The two reductions have different electron counts and sum to the target reaction.

Formula/Principle: Gibbs energies are additive; potentials themselves are not extensive.

Step 1:

$$\Delta G_1^\circ = -F(0.50), \quad \Delta G_2^\circ = -2F(-0.10) = +0.20F.$$

Step 2:

$$\Delta G_{sum}^\circ = -0.30F.$$

Step 3: The total electron count is three, so

$$E^\circ = -\Delta G_{sum}^\circ/(3F) = 0.10 \text{ V}.$$

Why: Electron weighting enters through the energy relation before converting back to the intensive potential.

Why not the others: (A) directly adds potentials. (B) takes an unweighted arithmetic mean. (D) reverses the sign of the final reduction.

Key point: Add energies for combined electrochemical reactions, then divide by the total electron count.

Q27 / D Excess-reactant control changes the observed order and slope without changing the intrinsic law

Answer: (D) Excess B produces a pseudo-first-order observation without changing the intrinsic two-reactant law.

Step 1: With B effectively constant in each run,

$$r = (k[B])[A] = k_{obs}[A].$$

Step 2: Integration gives $\ln[A] = \ln[A]_0 - k_{obs}t$. Thus ln[A] is linear in time.

Step 3:

$$k_{obs,Q}/k_{obs,P} = 0.40/0.20 = 2.$$

Why: The controlled concentration becomes part of the observed coefficient. A first-order trace in one monitored reactant is not proof that the intrinsic law contains only that reactant.

Why not the others: (A) ignores the excess-reactant approximation. (B) confuses constant intrinsic k with the concentration-dependent observed coefficient. (C) assigns an order from amount alone.

Key point: Distinguish an intrinsic rate constant from a pseudo-order coefficient.

Q28 / A A matched dilution blank changes the inferred weak-acid ionisation enthalpy

Answer: (A) HA ionisation is endothermic by 7.0 kJ mol⁻¹.

Given: Positive quoted heats mean energy released; the blank is a nonreaction contribution.

Formula/Principle: Hess's law compares the weak-acid reaction with the strong-ion neutralisation after subtracting the blank.

Step 1: Weak-acid reaction heat release alone is $5.20 - 0.20 = 5.00$ kJ. Its molar reaction enthalpy is -50.0 kJ mol⁻¹; the strong-acid value is -57.0 kJ mol⁻¹.

Step 2: Dissociation followed by proton neutralisation gives the weak-acid reaction.

$$\Delta H_{ion} + (-57.0) = -50.0 \;\Rightarrow\; \Delta H_{ion} = +7.0 \text{ kJ mol}^{-1}.$$

Why: Ionisation absorbs some of the heat released in the common proton–hydroxide step. A dilution background must not be assigned to that ionisation.

Why not the others: (B) omits the dilution correction. (C) reverses the enthalpy sign. (D) fails to convert the 0.100 mol heat difference to a per-mole value.

Key point: Correct measured heat contributions before using a Hess comparison.

Q29 / B Solid-phase exhaustion limits equilibrium gas pressure in one of two otherwise identical vessels

Answer: (B) P reaches 0.40 bar while Q exhausts carbonate at 0.10 bar.

Given: Each mole CO2 raises pressure by 1.00 bar in either vessel.

Formula/Principle: The Kp pressure condition for the displayed heterogeneous equilibrium requires both solid phases to remain available.

Step 1: Reaching 0.40 bar requires decomposition of 0.40 mol carbonate. P has enough, leaving 0.60 mol and retaining both solids.

Step 2: Q can release only 0.10 mol CO2 before its carbonate disappears.

$$p_Q = (0.10 \text{ mol})(1.00 \text{ bar mol}^{-1}) = 0.10 \text{ bar}.$$

Why: Once a required solid phase is absent, blindly setting the gas pressure to Kp would demand more reactant than exists. The phase-exhausted boundary must satisfy the material inventory.

Why not the others: (A) ignores Q's limiting carbonate amount. (C) equates all P carbonate with obligatory decomposition despite equilibrium. (D) reverses the inventories.

Key point: Check phase availability before imposing a heterogeneous equilibrium expression.

Q30 / C Formal f-electron accounting linked to lanthanoid exceptions and actinoid oxidation-state variation

Answer: (C) The formal counts are 4f0, 4f7 and 5f0, with electron energetics rather than atomic mass as the relevant explanation.

Step 1: Ce loses its two 6s, one 5d and one 4f valence electrons to reach Ce4+, leaving 4f0.

Step 2: Eu loses its two 6s electrons to form Eu2+, preserving the half-filled 4f7 arrangement.

Step 3: U6+ formally loses the six electrons distributed over 7s,6d and5f, leaving 5f0.

Why: The dominant +3 tendency of lanthanoids does not prohibit particular +2 or +4 states. Empty or half-filled configurations contribute to some exceptions, while participation of nearby-energy actinoid orbitals allows a wider range of oxidation states. This is not a prediction that every formally writable ion is equally stable in every medium.

Why not the others: (A) stops removing electrons despite the required charges. (B) has the counts but replaces orbital energetics with an unsupported mass rule. (D) gives cations extra electrons instead of removing them.

Key point: Reconcile charge with the whole valence configuration before explaining f-block exceptions.

Mathematics

Q31 / B Concrete matrix identity reduces a high-degree polynomial to a linear matrix expression

Answer: (B) -1.

Given: The concrete product identity can reduce every higher power.

Formula/Principle: Multiply the identity by powers of A and combine like matrix terms before taking a determinant.

Step 1: $A^3 = 3A^2 - 2A = 7A - 6I$, and $A^5 - 3A^4 = A^3(A^2 - 3A) = -2A^3$.

Step 2: Hence the matrix inside the determinant is

$$-2(7A - 6I) + 4(3A - 2I) - I = 3I - 2A.$$

Worked derivation:

$$3I - 2A = \begin{pmatrix} 3 & -2 \\ 4 & -3 \end{pmatrix}, \quad \det = -9 + 8 = -1.$$

Why: Matrix multiplication distributes over these polynomials in the same matrix. Reducing the polynomial is shorter and less error-prone than expanding fifth powers entry by entry.

Why not the others: (A) loses the determinant sign. (C) uses only one diagonal entry. (D) confuses a scalar multiple or trace with the determinant.

Key point: Reduce the matrix polynomial before evaluating its determinant.

Q32 / D Repeated intact blocks and disjoint endpoint cases in a multiset arrangement

Answer: (D) 12.

Given: The two AB tiles are indistinguishable from each other, as are the two C tiles.

Formula/Principle: Count the endpoint cases and divide by factorials for repeated units.

Step 1: Fix D at the left end. The four remaining tiles can be arranged in

$$4!/(2!2!) = 6$$

distinct ways.

Step 2: D at the right end gives another six. Since there is only one D tile, these two cases cannot overlap.

Worked derivation:

$$N = 6 + 6 = 12.$$

Why: AB is one indivisible tile, not two letters whose positions may be separated. Identical-tile permutations do not make different rows.

Why not the others: (A) counts only one end. (B) divides by only one repeated-pair factorial after selecting the endpoint. (C) counts unrestricted positions of D.

Key point: Identify the true repeated units before imposing an endpoint condition.

Q33 / A An onto composition constrains the final factor but leaves an unused intermediate element

Answer: (A) Twelve maps work; g is necessarily onto but f is not onto its four-element codomain.

Given: The composition uses g only at a,b,c.

Formula/Principle: An onto composition must attain both final outputs on the image of f.

Step 1: Of the eight assignments of g on a,b,c, exclude the two constant assignments. Six assignments remain.

Step 2: g(d) can independently be 0 or 1. Thus

$$N = (2^3 - 2)2 = 12.$$

Worked derivation: Both outputs already occur among g(a),g(b),g(c), so g is onto; f still never reaches d.

Why: Surjectivity of the composition forces the last map to cover its codomain, but does not require the first map to use every intermediate element.

Why not the others: (B) reverses the necessary implication. (C) subtracts too few constant-on-image assignments and misstates f's property. (D) omits the choice at d and contradicts g's forced surjectivity.

Key point: Distinguish the intermediate codomain from the actual image used by the composition.

Q34 / C Conjugate pairing and cubic root product determine a real but negative square sum

Answer: (C) -2.

Given: Nonreal roots of a real-coefficient polynomial occur in conjugate pairs.

Formula/Principle: Use conjugacy and the root product, then compute complex squares rather than squared moduli.

Step 1: Another root is 1-2i, and their product is 5. The three-root product is 10, so the third root is 2.

Step 2:

$$(1 + 2i)^2 = -3 + 4i, \quad (1 - 2i)^2 = -3 - 4i.$$

Worked derivation:

$$\sum r_j^2 = (-3 + 4i) + (-3 - 4i) + 4 = -2.$$

Why: Conjugacy makes the final sum real, but does not make each complex square positive. The constant coefficient fixes the otherwise unknown third root.

Why not the others: (A) loses the negative real contribution. (B) uses an unrelated root product. (D) sums squared moduli, 5+5+4, rather than the requested complex squares.

Key point: Do not replace complex squares by squared magnitudes.

Q35 / D A minimum of two curves and an absolute value require two different sets of breakpoints

Answer: (D) 16(√2-1)/3.

Given: The minimum can change formula, and its selected formula can also change sign.

Formula/Principle: Locate crossings before applying the absolute value; exploit even symmetry only after identifying the pieces.

Step 1: $x^2=2-x^2$ gives $x=\pm1$. On [0,1] the minimum is x^2; on [1,2] it is $2-x^2$, whose zero is $\sqrt{2}$.

Step 2: Therefore

$$I = 2\left[\int_0^1 x^2 dx + \int_1^{\sqrt{2}} (2 - x^2) dx + \int_{\sqrt{2}}^2 (x^2 - 2) dx\right].$$

Step 3: The three positive-half contributions are $1/3, (4\sqrt{2} - 5)/3, (4\sqrt{2} - 4)/3$.

Worked derivation:

$$I = 2(8\sqrt{2} - 8)/3 = 16(\sqrt{2} - 1)/3.$$

Why: Switching which curve is lower and taking its absolute value are separate operations with different breakpoints.

Why not the others: (A) evaluates the signed integral without the absolute value. (B) counts only the positive half-axis. (C) loses the constant endpoint contributions.

Key point: Find both comparison crossings and zeros of the selected function.

Q36 / B A direction parameter gives parallelism while an independent base-point parameter gives coincidence

Answer: (B) k=4 and b=2.

Given: Coincidence requires parallel directions and a common point.

Formula/Principle: Compare direction ratios first, then check the displacement between base points.

Step 1: The x and z components force the second direction to be twice the first. Its y component must then be 4, so k=4.

Step 2: The base-point displacement is (b-1,2,-1). It must equal λ(1,2,-1). The y and z components give λ=1 and therefore b=2.

Worked derivation:

$$(2,2,1) = (1,0,2) + (1,2,-1),$$

so L2 starts on L1 and has the same direction.

Why: Parallel direction vectors alone permit distinct lines; base-point alignment supplies the extra condition for coincidence.

Why not the others: (A) gives parallel but displaced lines. (C) fails the direction ratio. (D) gives an inconsistent y-direction sign.

Key point: Test direction and position separately.

Q37 / A An indeterminate implicit derivative is resolved through two explicit branches with a shared tangent

Answer: (A) Both branches have tangent y=0.

Given: The implicit derivative formula divides by y and is not valid at y=0 without further analysis.

Formula/Principle: Factor the defining equation into its real branches.

Step 1:

$$y^2 = x^4 \;\Rightarrow\; y = x^2 \quad \text{or} \quad y = -x^2.$$

Step 2: Their derivatives are 2x and -2x, each tending to zero at x=0.

Worked derivation: Both branches pass through (0,0), and both tangent equations are y=0.

Why: An indeterminate quotient from one rearranged implicit equation does not decide the geometric tangent. Explicit branch representations resolve the point directly.

Why not the others: (B) interprets division by zero as an infinite slope without checking the numerator. (C) assigns slopes not obtained from either branch. (D) contradicts direct substitution.

Key point: Resolve branches when a formal implicit derivative becomes indeterminate.

Q38 / C Pairing observations with affine reflections changes a pooled variance through mean separation

Answer: (C) Mean 4 and population variance 7.

Given: Each original observation contributes a pair x,8-x.

Formula/Principle: Compute paired deviations about the new mean, including the separation of the two group means.

Step 1: Each pair sums to 8, so the new mean is 4.

Step 2: Each pair's deviations from 4 are x-4 and its negative. Thus new variance is $E[(x-4)^2]$ over the old set.

Worked derivation:

$$E[(x-4)^2] = E[(x-2)^2] - 4E[x-2] + 4 = 3 + 0 + 4 = 7.$$

Why: Reflection preserves within-group variance, but merging groups with means 2 and 6 adds a between-group contribution. Simply duplicating the original set would not do that.

Why not the others: (A) omits group separation. (B) treats reflection as unchanged duplication. (D) uses only the reflected group's mean.

Key point: Account for both within-group dispersion and different group centres.

Q39 / D A floor-function squeeze controls a trigonometric quotient at infinity

Answer: (D) 1.

Given: For x>1, n=⌊x⌋ is positive and satisfies n≤x<n+1.

Formula/Principle: Bound the real-to-integer ratio, then use the standard sine limit.

Step 1:

$$1 \le x/n < 1 + 1/n,$$

so x/n tends to 1 despite the floor function's jumps.

Step 2: Rewrite

$$x \sin(1/n) = \frac{x}{n} \frac{\sin(1/n)}{1/n}.$$

Worked derivation: Both factors tend to 1 as x and n tend to infinity, giving limit 1.

Why: The relative size of each integer jump vanishes; discontinuity of the inner floor function does not by itself prevent this asymptotic limit.

Why not the others: (A) keeps the small sine factor but ignores the growing x factor. (B) infers failure from jumps without the squeeze estimate. (C) uses an incorrect limiting ratio x/n.

Key point: Control a discontinuous inner argument by a quantitative bound.

Q40 / A Different sampling protocols produce different conditional box probabilities from identical contents

Answer: (A) 3/5 for P and 3/4 for Q.

Given: The two protocols assign different prior weights to the box labels.

Formula/Principle: Apply Bayes's rule using the actual sampling mechanism.

Step 1: Under P,

$$P(B \mid R) = \frac{(1/2)(3/4)}{(1/2)(1/2) + (1/2)(3/4)} = 3/5.$$

Step 2: Under Q, the four red balls are equally likely conditional on red, and three belong to B.

Worked derivation:

$$P_Q(B \mid R) = 3/4.$$

Why: Uniform selection of boxes is not uniform selection of all balls when box sizes differ. Conditioning does not erase that initial difference.

Why not the others: (B) exchanges protocols. (C) ignores the red observation and the unequal box-size weighting. (D) updates P but not Q.

Key point: Specify the observation procedure before choosing a Bayes denominator.

Q41 / B A squared trigonometric factor splits connected solution bands without changing sign

Answer: (B) Three intervals with total length 3π/2.

Given: The inequality is strict and the range ends after a noninteger number of periods.

Formula/Principle: Factor before constructing the sign chart, retaining zeros even when a squared factor does not change sign.

Step 1:

$$\sin(2x)\cos x = 2\sin x\cos^2 x.$$

Positivity requires sin x>0 and cos x≠0.

Step 2: The intervals are $(0,\pi/2),(\pi/2,\pi),(2\pi,5\pi/2)$. The final endpoint is excluded because cos x=0.

Worked derivation:

$$N=3,\quad L=\pi/2+\pi/2+\pi/2=3\pi/2.$$

Why: The zero at π/2 splits a positive band without changing its sign; removing one point changes connectedness but not total interval length.

Why not the others: (A) ignores the interior zero that splits a band. (C) counts regions where sin x is negative. (D) omits the last positive partial period.

Key point: Track excluded zeros separately from sign changes.

Q42 / C A singular linear initial-value equation requires a component-specific global minimum

Answer: (C) The interval is (-∞,1) and the minimum is -1/8.

Given: The differential equation itself excludes x=1.

Formula/Principle: Solve with an integrating factor on the initial point's component, then minimise only on that component.

Step 1: The integrating factor is (x-1)², yielding

$$\frac{d}{dx}[(x-1)^2y]=1,\quad y=(x+C)/(x-1)^2.$$

Initial data give C=1.

Step 2: y=(x+1)/(x-1)² has a nonremovable singularity at 1, so its maximal initial-value interval is (-∞,1).

Step 3:

$$y'=\frac{-x-3}{(x-1)^3}.$$

On x<1 this is negative for x<-3 and positive for x>-3.

Worked derivation:

$$y_{min}=y(-3)=\frac{-2}{16}=-1/8.$$

Why: The coefficient singularity cannot be crossed merely by writing one algebraic expression. The sign chart and endpoint behaviour establish a global minimum on the correct interval.

Why not the others: (A) crosses the forbidden singularity. (B) overlooks negative values and the stationary minimum. (D) chooses the component not containing the initial point.

Key point: Set the maximal interval before making a global extremum claim.

Q43 / A Real-root discriminant and strict product positivity yield disconnected parameter intervals

Answer: (A) (1/2,1]∪[5,∞).

Given: Equal real roots are permitted, but zero product is not.

Formula/Principle: Combine discriminant nonnegativity with Vieta's root-product condition.

Step 1:

$$\Delta = (a+1)^2 - 4(2a-1) = a^2 - 6a + 5 = (a-1)(a-5) \geq 0.$$

Thus a≤1 or a≥5.

Step 2: The product is 2a-1>0, so a>1/2.

Worked derivation: Intersecting these conditions gives

$$(1/2, 1] \cup [5, \infty).$$

Why: The discriminant and product impose independent restrictions. The endpoints 1 and 5 stay because repeated roots are allowed; 1/2 is excluded by strict product positivity.

Why not the others: (B) uses the negative-discriminant region. (C) ignores the discriminant gap. (D) omits the positive-product restriction.

Key point: Preserve each inequality's endpoint convention during intersection.

Q44 / D An orthonormal change of coordinates converts a line equation without changing distance

Answer: (D) u=2 and distance 2.

Given: The coordinate transformation uses orthonormal direction vectors (3,4)/5 and (-4,3)/5.

Formula/Principle: An orthogonal rotation preserves lengths; alternatively use the original point-line distance formula.

Step 1:

$$3x + 4y = 10 \;\Rightarrow\; 5u = 10 \;\Rightarrow\; u = 2.$$

Step 2: In the rotated frame this line is parallel to the v-axis, two units from the origin.

Worked derivation:

$$d = |10|/\sqrt{3^2 + 4^2} = 2.$$

Why: Dividing by the normal's length makes u a distance coordinate; the raw linear form 3x+4y is not itself a unit-normal coordinate.

Why not the others: (A) exchanges the rotated axes. (B) omits the factor 5 in both equation and distance. (C) transforms the equation but fails to normalise the distance.

Key point: Normalise the rotated axis before interpreting its coordinate metrically.

Q45 / C An ordered triangle fixes the sign of a plane normal and a signed projection

Answer: (C) The oriented unit normal is (1,1,1)/√3 and the signed displacement is 2/√3.

Given: The ordered cross product fixes orientation, not merely a normal line.

Formula/Principle: Normalise AB×AC and project P-A onto that unit normal.

Step 1:

$$AB = (-1, 1, 0), \quad AC = (-1, 0, 1), \quad AB \times AC = (1, 1, 1).$$

Step 2: Its norm is √3, so n=(1,1,1)/√3.

Worked derivation:

$$(P - A) \cdot n = (0, 1, 1) \cdot (1, 1, 1)/\sqrt{3} = 2/\sqrt{3}.$$

Why: Reversing the vertex order would reverse the normal and signed distance while leaving the unsigned distance unchanged.

Why not the others: (A) gives a sign inconsistent with its normal. (B) reverses the prescribed orientation without changing the projected sign accordingly. (D) uses a nonunit normal as if it measured perpendicular distance.

Key point: Specify orientation and normalisation before reporting a signed distance.

Physics

Q46 / B A continuous piecewise velocity returns to its start after two reversals but travels a nonzero path

Answer: (B) Net displacement is zero and distance is 2 m.

Given: Velocity changes sign at t=1 and t=3, not at the continuous join t=2.

Formula/Principle: Displacement is signed area under v(t); distance is area under |v(t)|.

Step 1: Each two-second piece has equal positive and negative triangular areas, giving zero net displacement.

Step 2: There are four triangles of base 1 s and height 1 m s⁻¹.

Worked derivation:

$$s = 0, \qquad D = 4\left(\tfrac{1}{2} \cdot 1 \cdot 1\right) = 2 \text{ m}.$$

Why: A return to the original position does not erase travelled path. The two direction changes must be used when replacing v by |v|.

Why not the others: (A) treats signed cancellation as zero distance. (C) uses unsigned area for displacement. (D) uses rectangular rather than triangular areas.

Key point: Separate direction reversals from joins between formulas.

Q47 / C A rope-strength limit must be applied at the global maximum tension, with a separate tautness check

Answer: (C) The greatest speed at the stated point is √(6gL).

Given: Tension changes around the circle and must be checked globally.

Formula/Principle: Use energy to connect the stated point to the bottom, where tension is largest.

Step 1: The point is L(1-cos60)=L/2 above the bottom, so

$$v_b^2 = v^2 + gL.$$

Step 2: Bottom tension is

$$T_b = mv_b^2/L + mg = mv^2/L + 2mg.$$

Setting this equal to 8mg gives v²=6gL.

Step 3: At this value top speed squared is $v_b^2 - 4gL = 3gL$, giving positive top tension 2mg; the rope remains taut.

Worked derivation:

$$v_{max} = \sqrt{6gL}.$$

Why: Checking only the tension at 60 degrees misses the larger bottom speed and gravitational contribution there.

Why not the others: (A) applies the breaking limit locally at 60 degrees only. (B) omits the gravitational tension contribution. (D) is smaller than the permitted maximum.

Key point: Compare a material limit with the largest force over the complete motion.

Q48 / A Closed-path work fixes total dissipation but only bounds an individual potential difference

Answer: (A) Six joules are dissipated, and the potential difference lies between 4 and 10 J.

Given: Kinetic energy is unchanged on each quasistatic leg; nonnegative losses D1,D2 occur.

Formula/Principle: External work equals stored-potential increase plus dissipative loss.

Step 1: Let ΔU=UB-UA. Then

$$10 = \Delta U + D_1, \quad -4 = -\Delta U + D_2.$$

Step 2: Adding gives D1+D2=6 J. Nonnegativity requires ΔU≤10 and ΔU≥4.

Worked derivation:

$$Q_{diss} = 6\ \text{J}, \qquad 4 \le \Delta U \le 10\ \text{J}.$$

Why: A closed cycle fixes the sum of losses but does not determine their split between two paths. The stored-potential change on one leg therefore remains bounded rather than uniquely known.

Why not the others: (B) adds work magnitudes instead of signed work. (C) mistakes cycle dissipation for a one-leg potential difference. (D) treats a return to the same state as zero dissipated energy.

Key point: Use signed cycle work and passive-loss constraints separately.

Q49 / D An axle torque requires uphill static friction to couple rotation to uphill translation

Answer: (D) Acceleration is g/3 uphill and friction is 5Mg/6 uphill.

Given: A pure torque drives rotation; only friction can provide the additional uphill translational force.

Formula/Principle: Couple centre-of-mass force balance to torque balance and the rolling constraint.

Step 1: Take uphill acceleration positive and positive angular acceleration in the corresponding rolling direction. With f uphill,

$$Ma = f - Mg/2.$$

Step 2: Friction's torque opposes the applied driving torque:

$$\tfrac{1}{2}MR^2(a/R) = MgR - fR.$$

Step 3: Substitute f=Ma+Mg/2:

$$\tfrac{1}{2}Ma = Mg - Ma - Mg/2 \ \Rightarrow\ a = g/3.$$

Worked derivation:

$$f = Mg/3 + Mg/2 = 5Mg/6$$

uphill.

Why: Static friction need not oppose centre-of-mass motion; it opposes the slipping tendency generated here by the applied torque.

Why not the others: (A) omits rotational inertia and gives the wrong friction direction. (B) reverses friction and violates translation balance. (C) does not satisfy the two equations simultaneously.

Key point: Solve for friction's sign rather than assuming it opposes travel.

Q50 / B Radial and prograde impulses reach the same escape energy through different vector additions

Answer: (B) A radial increment vc is required, while the prograde tangential increment is (√2-1)vc.

Given: Initial velocity is tangential and escape is tested at the unchanged radius.

Formula/Principle: Escape requires final speed at least √2 vc; add velocities vectorially before evaluating kinetic energy.

Step 1: A radial increment u is perpendicular to the initial velocity. At threshold

$$v_c^2 + u^2 = 2v_c^2 \;\Rightarrow\; u = v_c.$$

Step 2: A prograde tangential increment w adds in the same direction:

$$(v_c + w)^2 = 2v_c^2 \;\Rightarrow\; w = (\sqrt{2} - 1)v_c.$$

Worked derivation:

$$(u_{min}, w_{min}) = (v_c, (\sqrt{2} - 1)v_c).$$

Why: The same escape-energy threshold can require different impulse magnitudes because of the cross term between the old and added velocity.

Why not the others: (A) reverses the geometries. (C) reports the required total speed as the added speed. (D) satisfies neither threshold.

Key point: An impulse changes a velocity vector, not a scalar speed in every direction.

Q51 / C Soap-film work counts two interfaces rather than one projected rectangle

Answer: (C) Work is 1.2×10⁻⁴ J and the two-surface film requires twice the single-interface work.

Given: A soap film has two liquid–air surfaces.

Formula/Principle: Surface-creation work is γ times the total increase in interfacial area.

Step 1: Each surface gains

$$\Delta A = L\Delta x = (0.10)(0.020) = 0.0020 \text{ m}^2.$$

Step 2:

$$W = 2\gamma\Delta A = 2(0.030)(0.0020) = 1.2 \times 10^{-4} \text{ J}.$$

Worked derivation:

$$W/W_{one\ interface} = 2.$$

Why: The geometric rectangle is one projected area, whereas the film creates that area on both faces.

Why not the others: (A) counts only one face. (B) calculates two faces but gives a one-face ratio. (D) counts the two-surface factor twice.

Key point: Count interfaces before multiplying area by surface tension.

Q52 / A A specified power-law compression heats the gas while releasing heat

Answer: (A) Q=-RT0/2: heat leaves even though the gas warms.

Given: This specified path is not the reversible adiabatic path for a monatomic gas.

Formula/Principle: Obtain final temperature from the path and ideal-gas law; integrate P dV for work.

Step 1: Since TV^(1/2) is constant, Tf=2T0 and

$$\Delta U = (3R/2)(T_f - T_0) = 3RT_0/2.$$

Step 2: Write P=P0(V0/V)^(3/2). Then

$$W_{by} = P_0 V_0^{3/2} \left[-2V^{-1/2}\right]_{V_0}^{V_0/4} = -2P_0V_0 = -2RT_0.$$

Worked derivation:

$$Q = \Delta U + W_{by} = 3RT_0/2 - 2RT_0 = -RT_0/2.$$

Why: Compression work can exceed the internal-energy increase, so some energy leaves as heat. A temperature rise alone does not establish heat input.

Why not the others: (B) reverses heat direction. (C) identifies heat with internal-energy change and omits work. (D) assumes every power-law compression is adiabatic.

Key point: Use the actual process path rather than inferring heat from temperature change.

Q53 / D Equal exterior charge data conceal different interior field work for conducting and volume-charged spheres

Answer: (D) The field does negative work -kQq/(2R) in the volume-charged sphere and zero in the conductor.

Given: k=1/(4πε0); the test charge does not perturb the source distributions.

Formula/Principle: Apply Gauss's law inside each sphere, then integrate field work along the inward path.

Step 1: Uniform volume charge gives

$$Q_{enc}(r) = Q(r/R)^3, \quad E(r) = kQr/R^3$$

radially outward.

Step 2:

$$W_{field} = q\int_R^0 E(r)dr = \frac{kQq}{R^3}\left[r^2/2\right]_R^0 = -\frac{kQq}{2R}.$$

Step 3: The conductor has E=0 throughout its interior, so its field work is zero.

Worked derivation:

$$(W_{volume}, W_{conductor}) = (-kQq/(2R), 0).$$

Why: Equal total charge and outer radius give equal exterior fields, but do not give equal interior fields or potential changes.

Why not the others: (A) reverses the inward-work sign. (B) exchanges the materials. (C) treats interior work as if both had a central point charge.

Key point: Identify the interior charge distribution before using a point-charge formula.

Q54 / B Partial dielectric overlap creates a fixed-charge energy gradient and inward force

Answer: (B) Force is Q²/(4C0L), pulling the dielectric inward.

Given: Charge is fixed; the two plate-area regions share the same voltage.

Formula/Principle: Add their capacitances in parallel and use F=-dU/dx with U=Q²/(2C).

Step 1:

$$C(x) = \frac{\varepsilon_0 b}{d}[(L-x) + 3x] = C_0(1 + 2x/L).$$

Step 2: At x=L/2, C=2C0 and dC/dx=2C0/L.

Step 3:

$$F = -\frac{d}{dx}\frac{Q^2}{2C} = \frac{Q^2}{2C^2}\frac{dC}{dx}.$$

Worked derivation:

$$F = \frac{Q^2}{2(2C_0)^2}\frac{2C_0}{L} = \frac{Q^2}{4C_0L} > 0.$$

Why: Inward insertion lowers fixed-charge electrostatic energy; no battery is required to supply the mechanical work. The approximation neglects corrections to capacitance, not the energy change with overlap.

Why not the others: (A) uses the zero-insertion magnitude and wrong direction. (C) loses a factor two. (D) confuses isolation with absence of stored energy.

Key point: Use the electrical constraint actually maintained during displacement.

Q55 / C A velocity selector fixes equal speeds before mass-to-charge and energy ratios are compared

Answer: (C) Radius ratio 2 and kinetic-energy ratio 4.

Given: The selector imposes the same speed, not the same accelerating potential or kinetic energy.

Formula/Principle: Balance qE=qvB in the selector, then use r=mv/(qBa) in the analyser.

Step 1: Both selected speeds are v=E/B, independent of mass and charge.

Step 2:

$$r_2/r_1 = \frac{4m/(2q)}{m/q} = 2.$$

Step 3:

$$K_2/K_1 = \frac{(4m)v^2/2}{mv^2/2} = 4.$$

Worked derivation:

$$(r_2/r_1, K_2/K_1) = (2, 4).$$

Why: Equal selector speed fixes the energy ratio by mass, whereas the magnetic radius depends on mass divided by charge.

Why not the others: (A) ignores mass-to-charge differences in radius. (B) uses a same-accelerating-voltage rule instead of velocity selection. (D) ignores the second beam's doubled charge in the radius.

Key point: Establish the beam-selection constraint before comparing analyser trajectories.

Q56 / B Translation changes flux only while loop overlap with a bounded field changes

Answer: (B) Current is zero for a/v<t<3a/v.

Given: Loop area and field strength are constant, but their overlap changes at entry and exit.

Formula/Principle: Induced emf is minus the rate of change of magnetic flux through the moving loop.

Step 1: The trailing edge enters after a/v. The leading edge reaches the far boundary after 3a/v. Between these times the entire loop remains in the same uniform field.

Step 2: Its flux magnitude is then Ba^2, independent of position.

Worked derivation:

$$\mathcal{E} = -d\Phi/dt = 0 \quad \text{for} \quad a/v < t < 3a/v.$$

Why: Motion alone does not guarantee changing flux. Entry and exit change overlap area; uniform translation while fully immersed does not.

Why not the others: (A) is the growing-overlap entry interval. (C) is the shrinking-overlap exit interval. (D) substitutes velocity for flux change.

Key point: Track the field-overlap area, not just the wire's speed.

Q57 / A Changing the surrounding medium removes total internal reflection without changing internal ray geometry

Answer: (A) Air gives total internal reflection; the liquid allows refraction with sin r=3√6/8.

Given: Normal entry leaves the internal direction unchanged in either surrounding medium.

Formula/Principle: At the exit, Snell law requires $n_g \sin i = n_{out} \sin r$; transmission requires the computed sine not exceed one.

Step 1: In air,

$$\sin r = \sqrt{3} \sin 45^\circ = \sqrt{6}/2 > 1,$$

so no refracted ray is possible and total internal reflection occurs.

Step 2: In the liquid,

$$\sin r = \frac{\sqrt{3}}{4/3} \frac{1}{\sqrt{2}} = \frac{3\sqrt{6}}{8} < 1.$$

Worked derivation: The same internal incidence lies above the air critical angle but below the glass–liquid critical angle.

Why: Total internal reflection depends on both refractive indices, not solely the prism geometry.

Why not the others: (B) ignores the changed outside index. (C) accepts a sine exceeding one. (D) fails to divide by the liquid index.

Key point: Recalculate the critical condition when the outside medium changes.

Q58 / D Time-averaged phase switching removes contrast unlike a constant phase offset

Answer: (D) The fluctuating pattern averages to 2I0 at both positions; a stable π offset instead shifts the fringes.

Given: Each phase state lasts half the averaging time and the two incident intensities are unchanged.

Formula/Principle: Equal coherent contributions give intensity $2I_0[1 + \cos\delta]$; average intensities over the specified phase states.

Step 1: At an original bright position, the two states give 4I0 and zero, averaging to 2I0. At an original dark position they give zero and 4I0, again averaging to 2I0.

Step 2: More generally, $\cos\delta + \cos(\delta + \pi) = 0$, so the interference term vanishes in the time average.

Worked derivation:

$$\langle I \rangle = \tfrac{1}{2}(4I_0 + 0) = 2I_0$$

at both selected positions.

Why: Loss of stable relative phase during averaging reduces contrast; a fixed phase change retains coherence and moves maxima.

Why not the others: (A) omits switching and misstates a fixed-phase effect. (B) describes one phase state rather than the average. (C) violates the summed background intensity and invents a wavelength change.

Key point: Distinguish a stable phase shift from a fluctuating phase that averages away interference.

Q59 / D Orbit circumference and matter wavelength jointly determine ionic charge and level energy

Answer: (D) n=4,Z=8 and energy -54.4 eV.

Given: The wavelength is a matter wavelength around the orbit, not an emitted photon's wavelength.

Formula/Principle: Apply the orbit standing-wave condition before the radius and energy relations.

Step 1:

$$n = 2\pi r/\lambda = 2\pi(2a_0)/(\pi a_0) = 4.$$

Step 2:

$$2a_0 = a_0(16/Z) \;\Rightarrow\; Z = 8.$$

Step 3:

$$E = -13.6(8^2/4^2) = -54.4 \text{ eV}.$$

Worked derivation: The circumference contains four wavelengths, and $n^2/Z = 2$ reproduces the supplied radius.

Why: Radius alone fixes only a ratio of quantum numbers; the independent matter-wave condition separates them.

Why not the others: (A) violates the circumference count. (B) uses the right n but fails the radius. (C) doubles n despite the measured wavelength.

Key point: Use independent quantisation constraints before evaluating the energy.

Q60 / A Conditional survival distinguishes a statistical half-life from an individual nuclear lifetime

Answer: (A) Both next-interval survival probabilities are 1/2; their actual decay order is not fixed.

Given: For a nucleus present at time zero, survival to t has probability S(t)=exp(-λt), and λT=ln 2.

Formula/Principle: Condition on the fact that the older nucleus is still present; do not count its earlier survival twice.

Step 1: For the older survivor,

$$P(\text{survive to } 4T \mid \text{survived to } 3T) = \frac{S(4T)}{S(3T)} = \frac{2^{-4}}{2^{-3}} = \frac{1}{2}.$$

Step 2: The newly produced nucleus survives its next T with probability

$$S(T) = e^{-\lambda T} = \frac{1}{2}.$$

Worked derivation: More generally,

$$\frac{S(t+T)}{S(t)} = e^{-\lambda T} = \frac{1}{2},$$

independent of its elapsed survival time t.

Why: A constant decay constant gives the same future decay probability per unit time to each nucleus that remains. Half-life characterises an ensemble probability, not a scheduled maximum life or an ageing countdown for an individual nucleus. Equal probabilities do not predict the realised decay order of these two nuclei.

Why not the others: (B) uses the unconditional probability of surviving all four half-lives instead of conditioning on the first three already survived. (C) invents an initial interval during which decay is impossible. (D) treats three half-lives as an individual lifetime limit despite the nonzero exponential survival probability.

Key point: Reset the probability calculation at the observation time for nuclei known to have survived.
